ACHRAF RAHIBE

UAV & AVIONICS SYSTEMS ENGINEER



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UAV and avionics engineer developing aerial, ground and maritime systems from architecture and integration through testing. Leads multidisciplinary electronics work and delivers autopilot configuration, diagnostics and technical documentation.

TECHNICAL SKILLS

CAD & Simulation
Onshape, AutoCAD Electrical; Gazebo/SITL; Gmsh mesh preparation.

Manufacturing
FDM 3D printing, airframe assembly, CNC preparation; BOMs and workshop planning.

Electronics & Embedded
STM32/ESP32/Arduino; Jetson/Raspberry Pi; CAN, UART, SPI, I2C, Ethernet; ESC/VESC, BMS; KiCad/Altium familiarity.

Programming & Robotics
C/C++, Python, VHDL, Linux, Git; ArduPilot, PX4, MAVLink; Mission Planner/QGC; PID, FFT and logs.

Vision & Automation
YOLOv8/ByteTrack, RTSP/GStreamer, gimbal interfaces; Siemens PLC, TIA Portal and WinCC.

Design & Communication
Photoshop, Illustrator; engineering reports, automotive service documentation and technical coordination.

EDUCATION

State Engineer Diploma
Electrical Engineering and Telecommunications; Industrial Control.
FSTM, Hassan II University.
2021-2025 | Bac+5

DUT in Electrical Engineering
Electronics and Industrial Computing.
EST Fès, Université Sidi Mohamed Ben Abdellah.
2019-2021

Technical Baccalaureate
Electrical Sciences and Technology.
Lycée Technique Anoual, Casablanca.
2018-2019

LEADERSHIP

ROBOTHINGS: Founder/President (2023-2025). ENACTUS FSTM: Marketing Lead (2022-2023).

LANGUAGES

Arabic: Native · German: Basic
English & French: Fluent

PROFESSIONAL EXPERIENCE

FARASHA SYSTEMS | Rabat, Morocco

Electronics Team Manager | Mar 2026 - Present

- Lead avionics architecture, component selection and integration across electronics, embedded, RF and mechanical teams. Plan validation, resolve field issues and maintain technical documentation.

UAV & Avionics Systems Engineer | Aug 2025 - Mar 2026

- Integrated UAV and USV platforms, configuring autopilots, navigation, payloads and power interfaces. Supported simulation, bench checks, flight trials and water tests.

Independent engineering | Upwork, Fiverr and direct clients

UAV & Robotics Engineer | 2025 - Present

- Deliver autopilot setup, tuning and log-based troubleshooting. Support on-site teams remotely and turn findings into corrective actions, reports and practical test plans.

CAPGEMINI ENGINEERING | Casablanca

Technical Documentation Technician / Consultant | Jul - Sep 2023

- Supported Stellantis after-sales engineering by translating vehicle-system data into structured diagnostic and service documentation.

ENGINEERING PROJECTS

VTOL & fixed-wing aircraft
ArduPilot / PX4 · Cube / Pixhawk · CAD / 3D printing

- Integrated airframes, propulsion, avionics and telemetry; configured flight modes and missions. Supported hover, transition and stability testing, including successful client VTOL trials.

Unmanned surface vehicle (USV)
ArduRover · Differential thrust · Waterjets · Jetson Nano

- Developed the full system from architecture and electrical integration through water tests. Integrated autopilot navigation, differential-thrust waterjets, FPV and AI detection with Jetson Nano and an A8C gimbal camera.

Rotorcraft control & diagnostics
CUAV / Matek · PID tuning · FFT / notch filtering

- Configured 17- and 30-inch quadcopters and investigated unstable flight through logs. Reviewed helicopter control, vibration and estimator behaviour; documented corrective actions and subsequent tests.

Ground robotics & UAV docking
CUAV X7 · Dual VESC · ArduRover · MAVLink

- Integrated rover motor-control interfaces and developed a four-actuator docking proof of concept with synchronized control, isolated power and autopilot handshakes.

Embedded vision & avionics
Jetson / Raspberry Pi · YOLOv8 / ByteTrack · RTSP

- Integrated camera, tracking and video interfaces; reviewed companion-computer baseboard connectivity. Prepared AprilTag precision-landing architecture and staged verification criteria.